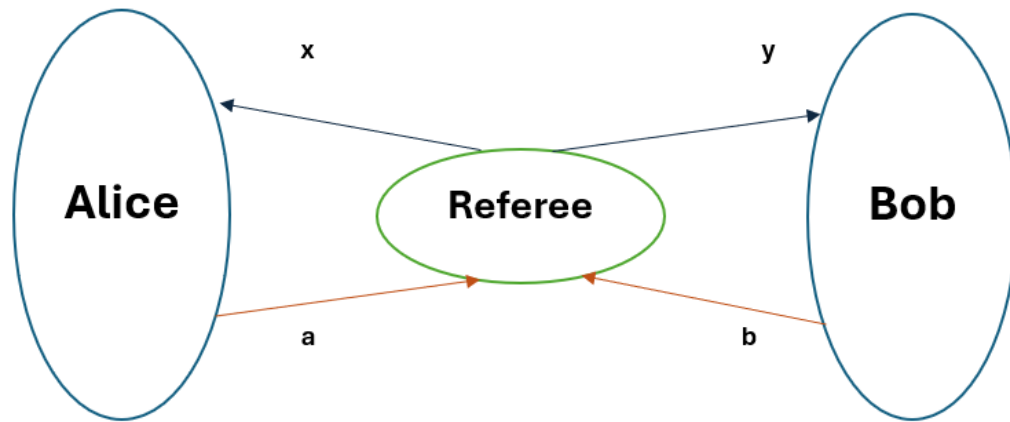


Open Problems in Multipartite Nonlocality

Hafiza Rumla Amer

Bipartite Correlations



Goal: $a \oplus b = x \wedge y$

1. Alice and Bob are spatially separated without means to communicate.
2. Both receive inputs $x, y \in \{0,1\}$ and produce outputs $a, b \in \{0,1\}$.
3. Goal is to compute the AND function successfully.
4. The correlation is represented by a joint conditional probability distribution $P(ab|xy)$.

Bipartite Strategies

- Deterministic (Local) Strategies

Alice and Bob meet beforehand and agree upon some deterministic strategy. The maximum winning probability is $p_{win} = 0.75$.

Distribution:

$$p(ab|xy) = \sum_{\lambda} q_{\lambda} p_{\lambda}(a|x) p_{\lambda}(b|y)$$

Expected Value (CHSH Inequality):

$$V = \langle a_0 b_0 \rangle + \langle a_0 b_1 \rangle + \langle a_1 b_0 \rangle - \langle a_1 b_1 \rangle \leq 2$$

Where $V = \sum_{x,y} (-1)^{x \wedge y} \langle a_x b_y \rangle$ and $a_x, b_y \in \{+1, -1\}$.

Nonlocal Correlations

No-Signaling Constraint:

$$\begin{aligned} p(a) &= \sum_b p(ab|xy) = \sum_b p(ab|xy') \\ p(b) &= \sum_a p(ab|xy) = \sum_a p(ab|x'y) \end{aligned}$$

Nonlocal Correlations:

$$p(ab|xy) \neq \sum_{\lambda} q_{\lambda} p_{\lambda}(a|x) p_{\lambda}(b|y)$$

Nonlocal Correlations

- Quantum Resource:

Alice and Bob use quantum entangled state to produce their outputs.

Maximum winning probability is $p_{win} = 0.85$.

Tsirelson's Inequality: $V = \langle a_0 b_0 \rangle + \langle a_0 b_1 \rangle + \langle a_1 b_0 \rangle - \langle a_1 b_1 \rangle \leq 2\sqrt{2}$

However, the algebraic maximum value is 4...

Nonlocal Correlations

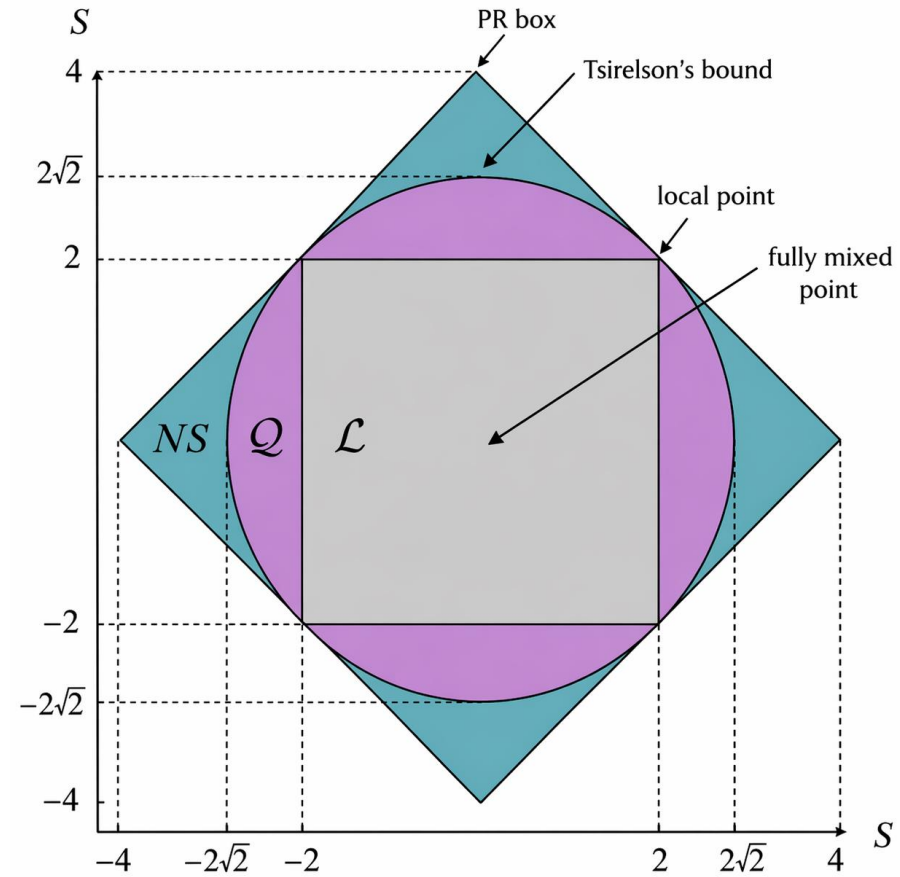
- Nonlocal Box as Resource:

Alice and Bob each has access to a nonlocal box to produce their outputs.

PR box achieves the maximum expected value; $V = 4$ and $p_{win} = 1$.

$$P(ab|xy) = \begin{matrix} & & \begin{matrix} ab \end{matrix} \\ & & \begin{matrix} 00 & 01 & 10 & 11 \end{matrix} \\ \begin{matrix} xy \\ 00 \\ 01 \\ 10 \\ 11 \end{matrix} & \left(\begin{matrix} 0.5 & 0 & 0 & 0.5 \\ 0.5 & 0 & 0 & 0.5 \\ 0.5 & 0 & 0 & 0.5 \\ 0 & 0.5 & 0.5 & 0 \end{matrix} \right) \end{matrix}$$

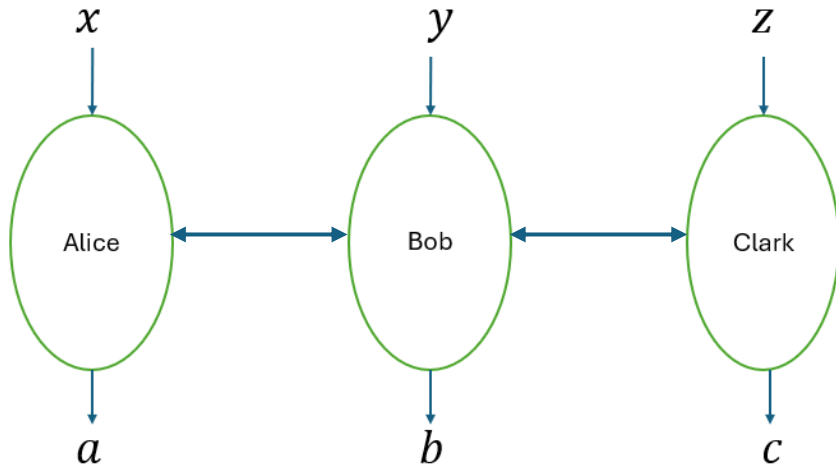
Nonlocal Correlations



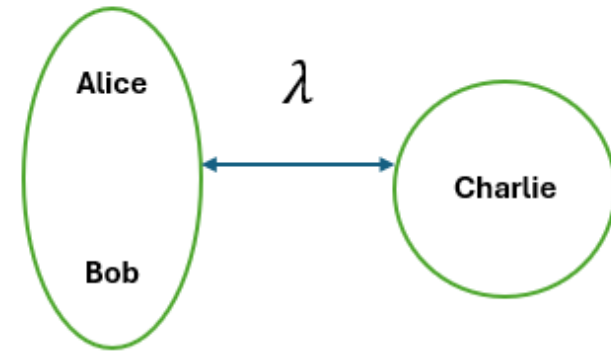
Genuine Tripartite Nonlocal Correlations

$$p(abc|xyz) \neq \sum_{\lambda} q_{\lambda} p_{\lambda}(ab|xy) p_{\lambda}(c|z) + \sum_{\mu} q_{\mu} p_{\mu}(bc|yz) p_{\mu}(a|x) + \sum_{\alpha} q_{\alpha} p_{\alpha}(ac|xz) p_{\alpha}(b|y)$$

Genuine Tripartite Nonlocal



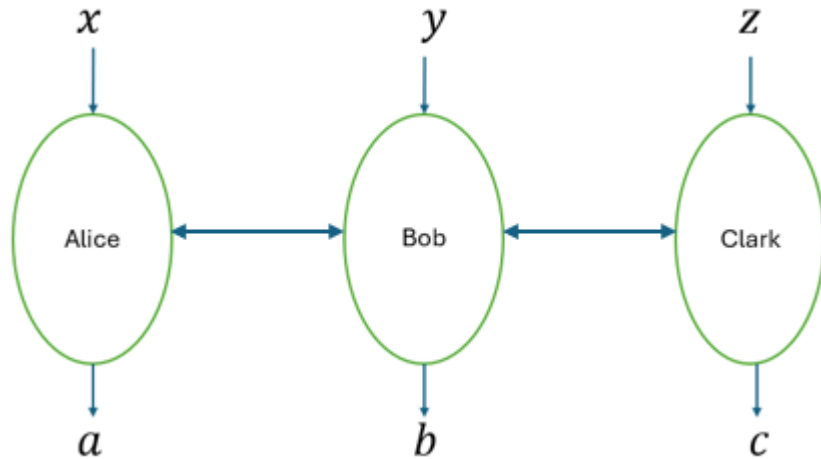
Svetlichny Framework



Simulating Tripartite Nonlocal Correlations

$$p(abc|xyz) \neq \sum_{\lambda} q_{\lambda} p_{\lambda}(ab|xy) p_{\lambda}(c|z) + \sum_{\mu} q_{\mu} p_{\mu}(bc|yz) p_{\mu}(a|x) + \sum_{\alpha} q_{\alpha} p_{\alpha}(ac|xz) p_{\alpha}(b|y)$$

Problem Statement: Is it possible to reproduce the target tripartite correlations exactly using only bipartite nonlocal boxes shared pairwise (Svetlichny Framework)?



Simulation by distribution:

$\min \epsilon$

$$\sum_i p_i N_i + \epsilon \text{Noise} = N$$

where,

N_i are bipartite NLB

N is a perfect tripartite NLB

Known Results:

Jonathan Barrett and Stefano Pironio, “**Popescu-Rohrlich correlations as a unit of nonlocality**”, arXiv:quant-ph/0506180

N. Brunner and P. Skrzypczyk, “**Bipartite Units of Nonlocality**,” *Physical Review Letters* **102**, 160403 (2009).

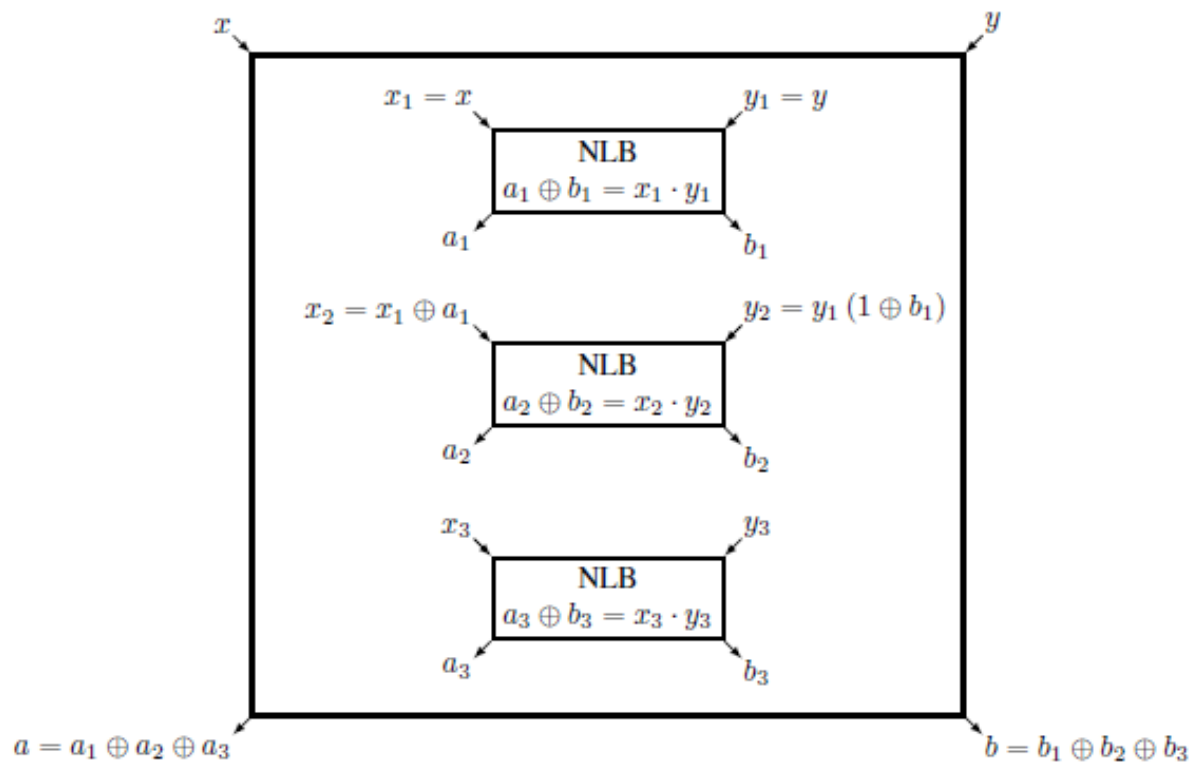
Multipartite Nonlocal Correlations

- Generalize the number of players to any arbitrary number n .
- Given any n partite XOR Nonlocal Box: $\bigoplus a_i = f(x_1, x_2, \dots, x_n)$ with binary inputs a_i and binary outputs x_i . How well the players can simulate it by using at most $n - 1$ partite nonlocal boxes? (Symmetric Boxes, Threshold Boxes)
- If the isotropic noise of n partite NLB is represented as ϵ , then which set of finite at most $n - 1$ NLBs will provide a lower bound on the error?

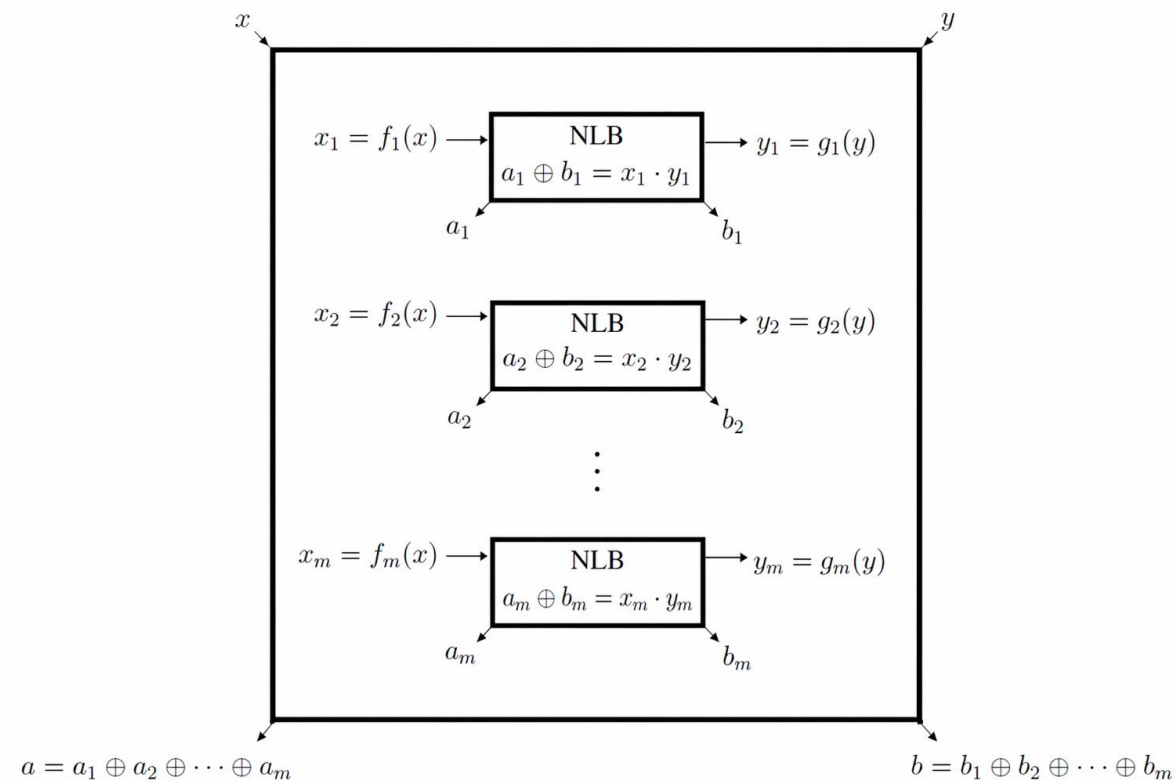
$$\sum_i p_i N_i + \epsilon \text{Noise} = N$$

- Can the non-adaptive protocols outperform adaptive protocols?

Adaptive Protocol

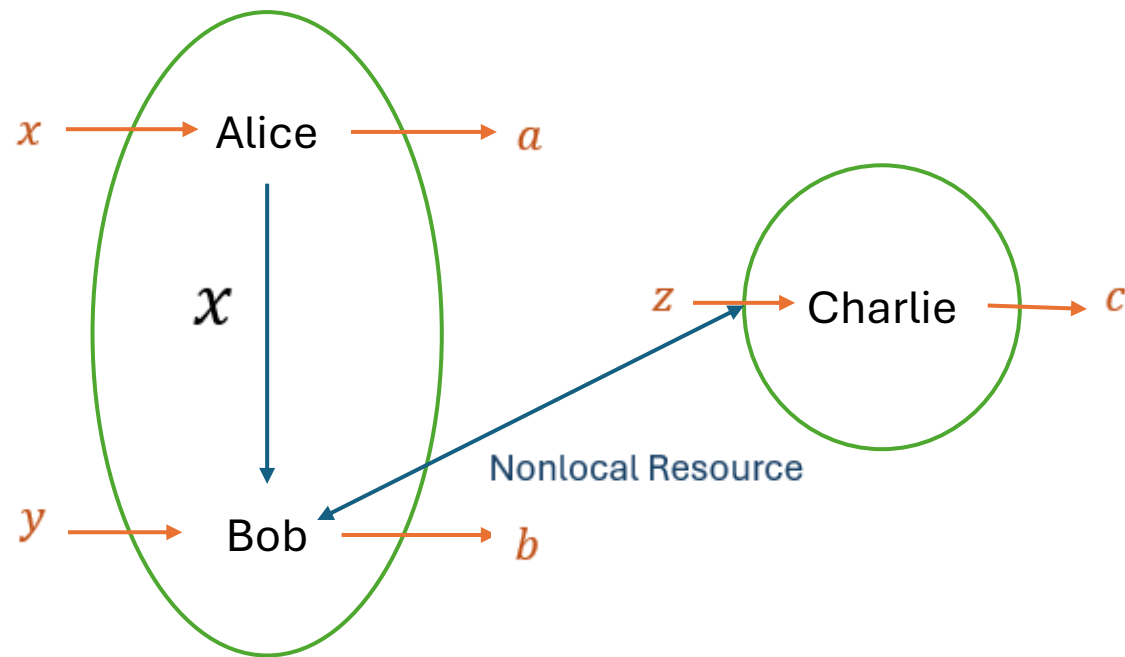


Non-Adaptive Protocol



LOCCG (Local Operations & Classical Communication)

$$p(abc|xyz) = \sum_{\lambda} q_{\lambda} p_{\lambda}(ab|xyz) p_{\lambda}(c|z)$$



Bob and Charlie share a nonlocal resource.

Alice sends her input bit x to Bob.

Alice outputs a deterministically.

Nonlocal Resources could be entanglement or NLB.

Open Problem:

- Results are available for Svetlichny Framework in different fields like cryptography, computational tasks, Optimization etc.
- Compare the results in Svetlichny framework to LOCCG framework and do the analysis.

Relevant Research Papers

- J. S. Bell. **On the Einstein–Podolsky–Rosen paradox**. *Physics*, 1:195, 1964.
- George Svetlichny. **Distinguishing three-body from two-body nonseparability by a Bell-type inequality**. *Phys. Rev. D*, 35:3066–3069, May 1987.
- S. Popescu and D. Rohrlich. **Quantum nonlocality as an axiom**. *Foundations of Physics*, 24:3, 1994.
- J. Barrett, N. Linden, S. Massar, S. Pironio, S. Popescu, and S. Roberts. **Nonlocal correlations as an information theoretic resource**. *Physical Review A*, 71:022101, 2005.
- J. Barrett and S. Pironio. **Popescu-rohrlich correlations as a unit of nonlocality**. *Physical Review Letters*, 95:140401, 2005.
- Jonathan Barrett and Stefano Pironio, “**Popescu-Rohrlich correlations as a unit of nonlocality**”, arXiv:quant-ph/0506180
- N. Brunner and P. Skrzypczyk, “**Bipartite Units of Nonlocality**,” *Physical Review Letters* **102**, 160403 (2009).
- Brunner et al., “**Bell Nonlocality**,” *Rev. Mod. Phys.* 86, 419 (2014)